

OS LAB

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Make Grade:

```
el.sym
make[1]: Leaving directory '/home/ibrahim/xv6-labs-2025'
== Test sleep, no arguments ==
$ make qemu-gdb
sleep, no arguments: OK (6.8s)
== Test sleep, returns ==
$ make qemu-gdb
sleep, returns: OK (1.1s)
== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: OK (1.2s)
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (1.1s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (2.7s)
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: OK (2.4s)
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (1.2s)
== Test memdump, format i, s, p ==
$ make qemu-gdb
memdump, format i, s, p: OK (1.4s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.5s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.6s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (2.7s)
== Test exec ==
$ make qemu-gdb
exec: OK (1.7s)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: OK (2.8s)
== Test exec, recursive find ==
$ make qemu-gdb
Timeout! exec, recursive find: OK (30.3s)
== Test time ==
time: FAIL
Cannot read time.txt
Scores: 120/131
make: *** [Makefile:368: grade] Error 1
ibrahim@DESKTOP-T6NVIN2:~/xv6-labs-2025$
```

Sleep:

```
exec [B failed
$ sleep 100
|
```

Sixfive:

```
ibrahim@DESKTOP-T6NVIN2:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt
ep user/_init user/_kill user/_ln user/_ls user/_mkdir user/
s user/_grind user/_wc user/_zombie user/_logstress user/_
ive user/_uptime user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bit
ballocc: first 1117 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kern
irtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sixfive sixfive.txt
5
100
18
6
$ |
```

Memdump:

```
$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
B80
1819438967
100
z
xyzyz
Example 5:
hello
w
o
r
l
d
```

Find:

```
$ echo > a/b
$ find . b
exec find. failed
$ find . b
./b
./a/b
$ |
```

Exec:

```
$ find . wc -exec echo hi
hi ./wc
$ |
```

Optional

Uptime:

```
$ uptime
uptime in ticks: 6396 ticks
```

Modified shell:

```
$ ls
s.          1 1 1024
..          1 1 1024
README     2 2 2425
findtest.sh 2 3 91
sixfive.txt 2 4 19
cat         2 5 36512
echo        2 6 35368
forktest    2 7 17080
grep        2 8 39952
init         2 9 35808
kill         2 10 35288
ln           2 11 35096
ls           2 12 38648
mkdir       2 13 35344
rm           2 14 35336
sh           2 15 65696
stressfs    2 16 36216
usertests   2 17 186256
grind        2 18 51744
wc           2 19 37496
zombie       2 20 34688
logstress   2 21 37288
forphan      2 22 36112
dorphan      2 23 35552
sleep        2 24 35176
sixfive      2 25 36568
uptime       2 26 35288
find         2 27 42024
mempdump     2 28 38800
console      3 29 0
b            2 30 0
a            1 31 48
$ history
1: echo > b
2: mkdir a
3: ls
4: echo > a/b
5: find . b
6: find . b
7: find . wc -exec echo hi
8: uptime
9: ls
10: history
$ |
```